

Sequences and Series

Useful formulas:

$$1. \quad 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$2. \quad 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$3. \quad 1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

1. Compute $1 + 2 + 3 + \dots + 100$.
2. Compute $1 + 3 + 5 + \dots + 99$.
3. Compute $2 + 4 + 6 + \dots + 200$.
4. Compute $1^2 + 2^2 + 3^2 + \dots + 100^2$.
5. Compute $1^3 + 2^3 + 3^3 + \dots + 10^3$.
6. Compute $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + 10 \cdot 11$.
7. Compute $1^2 + 3^2 + 5^2 + \dots + 9^2$.
8. Compute $1^3 + 3^3 + 5^3 + \dots + 9^3$.
9. Compute $1 - 2 + 3 - 4 + \dots + 99 - 100$.
10. Find $1 \cdot 2 - 2 \cdot 3 + 3 \cdot 4 - \dots + 99 \cdot 100 - 100 \cdot 101$.
11. Find $1 \cdot 5 + 2 \cdot 6 + 3 \cdot 7 + \dots + 98 \cdot 102$.
12. Find the sum $1 \cdot 3 + 2 \cdot 6 + 3 \cdot 11 + \dots + 97 \cdot (97^2 + 2)$.
13. Compute
$$1 \cdot 2^2 + 2 \cdot 3^2 + 3 \cdot 4^2 + \dots + 99 \cdot 100^2.$$
14. The n th term of a sequence is $a_n = n^2 - n$. Find $\sum_{n=1}^{20} a_n$.
15. Compute the alternating sum

$$1^2 - 2^2 + 3^2 - 4^2 + \dots + (2n-1)^2 - (2n)^2.$$

Express your answer in terms of n .

16. Compute

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{99 \cdot 100}.$$

17. Compute

$$\sum_{k=1}^{20} \frac{1}{k(k+2)}.$$

18. Compute

$$\sum_{k=1}^{15} k(k+1)(k+2).$$

19. Compute

$$1^3 - 2^3 + 3^3 - 4^3 + \cdots + 99^3 - 100^3.$$

20. The sum of the first n terms of a series is $S_n = 5n^3 - 3n + 1$. Find the 100th term of the series.

21. Let $a_1, a_2, \dots$ be an arithmetic sequence. If $a_1 + a_2 + \cdots + a_{100} = 50$ and $a_{101} + a_{102} + \cdots + a_{200} = 100$, find $a_{201} + a_{202} + \cdots + a_{300}$.

22. Let $a_1, a_2, \dots$ be a geometric sequence. If $a_1 + a_2 + \cdots + a_{100} = 50$ and $a_{101} + a_{102} + \cdots + a_{200} = 100$, find $a_{201} + a_{202} + \cdots + a_{300}$.

23. The sum of the first n terms of a series is $S_n = 3^n - 2^n$. If the first term of the series is a_1 and the n th term is a_n , find the alternating sum $a_1 - a_2 + a_3 - \cdots + a_{99}$.

24. Find the sum

$$\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots + \frac{100}{2^{100}}$$

25. Compute

$$\sum_{k=1}^n k \cdot 2^k.$$

26. Compute

$$1 + 2x + 3x^2 + 4x^3 + \cdots + nx^{n-1}$$

for $x \neq 1$. (Express as a closed form.)

27. Let $S = 1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + n \cdot n!$. Find a closed form for S .

28. Find the sum of all integers from 1 to 200 that are divisible by 3 or 5.

29. The sum of n consecutive integers starting at a equals 1000. Find all pairs (a, n) with $n > 1$ and $a \geq 1$.

30. Find the value of

$$(1 - 2^2)(1 - 3^2)(1 - 4^2) \cdots (1 - 10^2).$$

31. Compute

$$\frac{1^2}{1 \cdot 3} + \frac{2^2}{3 \cdot 5} + \frac{3^2}{5 \cdot 7} + \cdots + \frac{n^2}{(2n-1)(2n+1)}.$$

32. Find the sum

$$\sum_{k=1}^{100} \frac{k}{(k+1)!}.$$

33. Let $f(n) = 1^2 + 2^2 + \cdots + n^2$. For how many integers $1 \leq n \leq 100$ is $f(n)$ divisible by $2n+1$?

34. Compute

$$\sum_{k=1}^n \sum_{j=1}^k j.$$

Express as a closed form in n .

35. Find the sum

$$\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{99} + \sqrt{100}}.$$